

Hynetd v0.2.3 Manual

(Hybrid Network Topology Discovery)

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1 Required libraries

Hynetd requires these libraries to compile and work correctly:

- **net-snmp** >= 5.2
- **libpthread**

2 License

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3 Installation

The easiest way to compile hynetd is launching the following command:

```
bash$ make hynetd
```

read INSTALL for further details...

4 How to use it

First of all, root privileges are needed to use this tool, because it uses standard raw sockets to manage ICMP packets.

Hynetd also needs at least 3 parameters to be defined on command line:

-t <max_number_of_threads>	number of threads to be created during scan
<base_address>	base IP address of the first range
<address_space_size>	amplitude of the first range

The correct syntax to be used can be displayed using '-h' option.

```
HyNetD [-h] | [--help] |  
        [-v] [-d] [-a] [-r[G|S|A] <retries_number>] [-T[G|S|A] <timeout>] [-L <max_ttl>] [-I]  
        [-c <community_name> [[community_name] ... ]] -t <max_number_of_threads>  
        <base_address> <address_space_size> [[<base_address> <address_space_size>] ... ]
```

NOTE: It's important that -t option is specified as the last '-' option.

Here are described all other options:

-d enable DNS inverse name look-up (default = disabled)

With this option enabled, hostnames are retrieved using DNS inverse look-up queries. This way, during alias resolution phase, addresses resolved to the same hostname are considered aliases.

-c community_name list to be used in SNMP requests (default = public)

Community names defined here are used in order to test if SNMP is enabled on each ping responsive ip address. There is no size limit to this list, so a wordlist can be used in a command like this: `hynetd -c `cat wordlist` -t . . .`

-a enable ARP table extraction (default = disabled)

ARP tables are extracted using SNMP and shown in the results. This option can be useful for future features.

-r|-rG number of retries in scan process packet sending (default = 1)

This is the number of retries used for all the scan process operations (ping, pingRR, backtrace ...). Different values are used only during Ally algorithm and snmpTest function.

-rS number of retries in snmpTest packet sending (default = 1)

This is the number of retries used during snmpTest. It can be set to an higher value than scan process retries to maintain good performance without missing SNMP enabled routers.

-rA number of retries in Ally algorithm packet sending (default = 1)

This is the number of retries used during Ally algorithm. It can be set to an higher value to improve alias resolution results.

-T -TG timeout value used during scan process in ms (default = 500 ms)

This is the timeout value used during all operations done in scan process (ping, pingRR, backtrace ...). Different values are used only during Ally algorithm and snmpTest function.

-TS timeout value used during snmpTest function in ms (default = 500 ms)

This is the timeout value used during snmpTest. It can be set to an higher value than scan process timeout to maintain good performance without missing SNMP enabled routers.

-TA timeout value used during Ally algorithm in ms (default = 1000 ms)

This is the timeout value used during Ally algorithm. It can be set to an higher value to achieve better alias resolution results. Values lower than 1000ms are not recommended, because most routers have a rate limit of 1 packet per second for ICMP_PORT_UNREACH replies.

-L maximum value of ttl to use during backtrace algorithm (default = 20)

This is the maximum ttl value used to reach destination host during backtrace algorithm. Setting this parameter to a very high value can cause performance loss. A very low value can avoid some hosts to be reached.

-I set backtrace mode to ICMP (default = UDP)

If enabled, backtrace algorithm starts its process using ICMP packets instead of UDP packets. It can avoid the generation of extra traffic when UDP packets are filtered on most routers.

-v enable verbose mode

If enabled, the output shown is more verbose.

5 Examples

In this section there are some examples that show common usages and syntax of Hynetd.

5.1 Example 1

In this example we scan 2 ranges with some options defined:

```
# hynetd -c private public -t 50 192.168.1.1 80 192.168.15.20 50
```

In this case hynetd scan addresses from **192.168.1.1** to **192.168.1.80** and **192.168.15.20** to **192.168.15.70** using 50 threads and two community names.